



Project Based Learning –I (24CFT0202)

Team Details:

Team Name : **SmartHire**

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What is SmartHire AI?

SmartHire is an end-to-end recruitment platform designed to eliminate human bias and operational friction through local AI inference



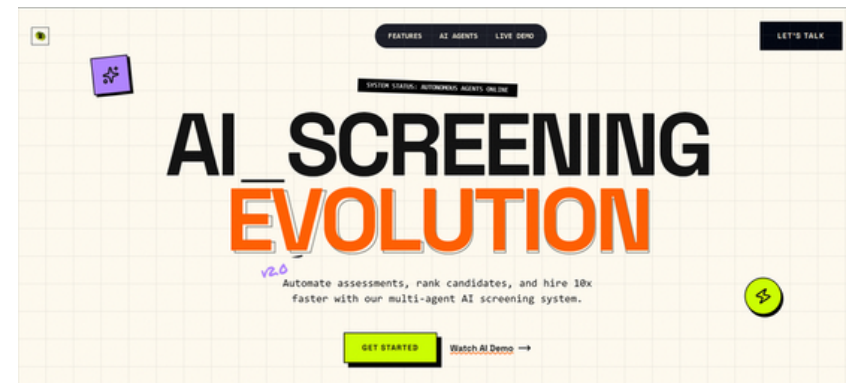
Automated Screening: Parses and ranks resumes using semantic vectors.



AI Interviewer: Conducts real-time technical & behavioral assessments.

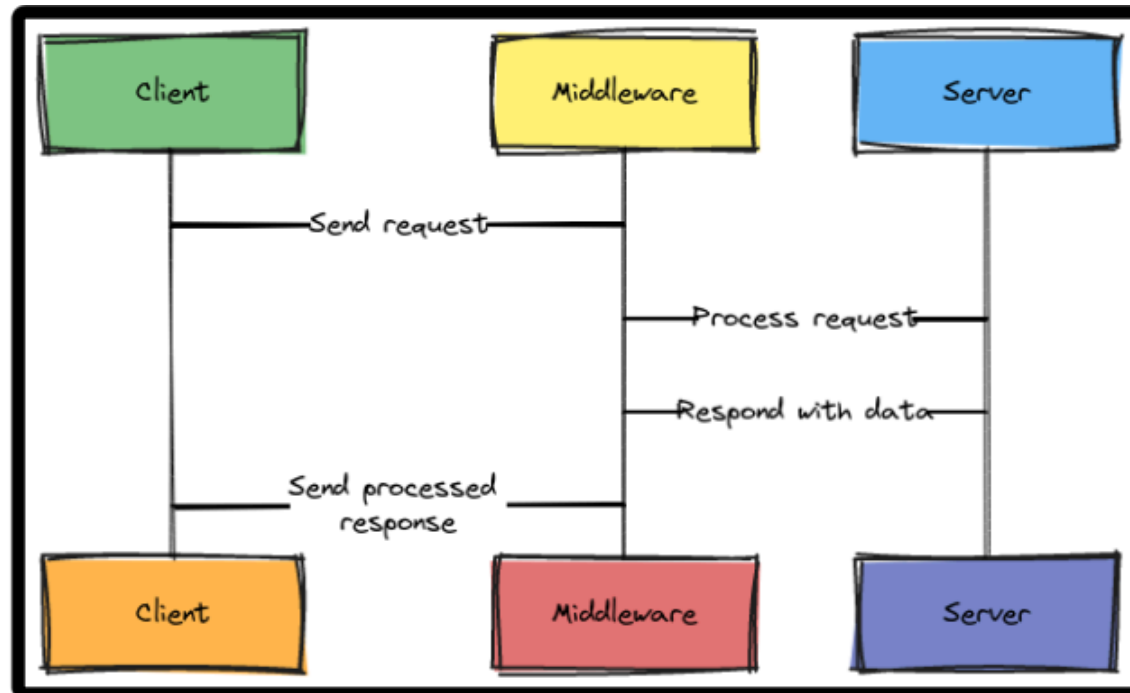


Integrity First: Computer vision ensures proctored environment.



The Challenge in Hiring

- **Manual Fatigue:** High volume of resumes leads to candidate fatigue and missed talent.
- **Unconscious Bias:** Human reviewers often favor specific keywords or formatting over actual skill.
- **Integrity Gaps:** Remote hiring lacks robust anti-cheating mechanisms for technical tests.
- **AI Costs:** SaaS-based LLM APIs are expensive and risk data privacy leaks.



Advanced Tech Stack

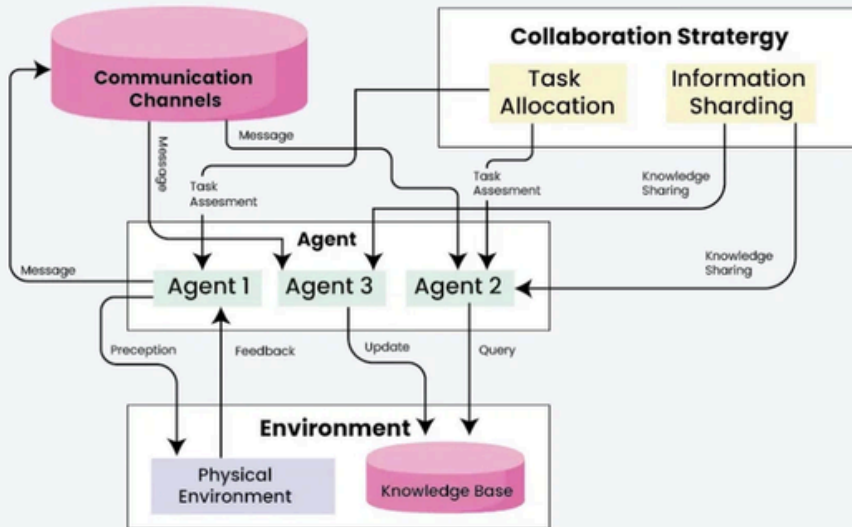


Area	Stack Details
Frontend	React • Vite • TypeScript • TailwindCSS • shadcn/ui • React Router DOM
Backend	FastAPI • SQLAlchemy • SQLite
AI & Vector Search	Ollama • qwen2.5-coder:7b • gemma3:4b • ChromaDB • CrossEncoder
Security & Monitoring	OpenCV • MediaPipe • YOLOv8 • Docker Sandbox
Deployment	Vercel • Ngrok

Collaborative AI Multi-Agents

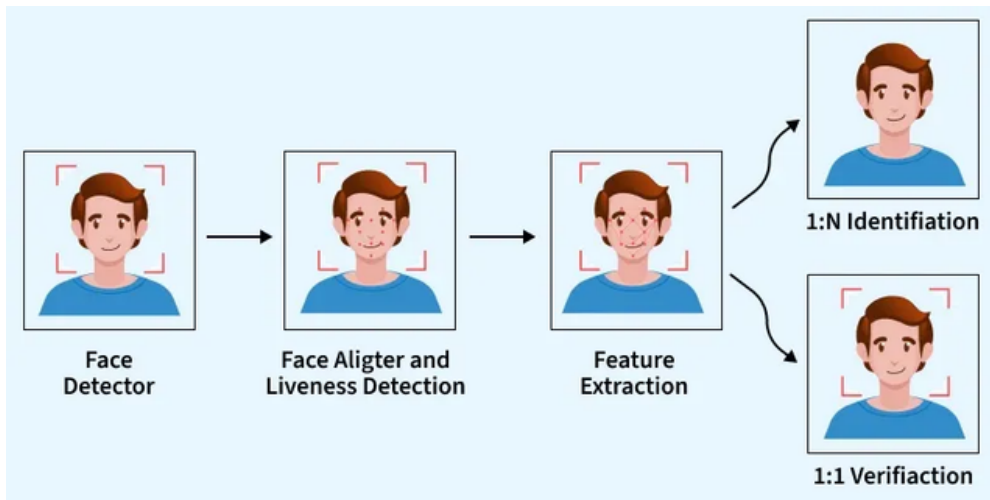
Our system utilizes five distinct specialized agents that collaborate to provide a holistic candidate score.

Multiagent system Architecture



Agent	Model Target	Role
Interviewer	Qwen2.5-coder	Dynamic Technical Interview
Evaluator	Qwen2.5-coder	Behavioral & Soft Skills
Validator	CrossEncoder	Semantic JD Matching
Assessor	Gemma3:4b	DSA Challenge Generator
Final Judge	Qwen2.5-coder	Consolidated JSON Report

Computer Vision Guardrails

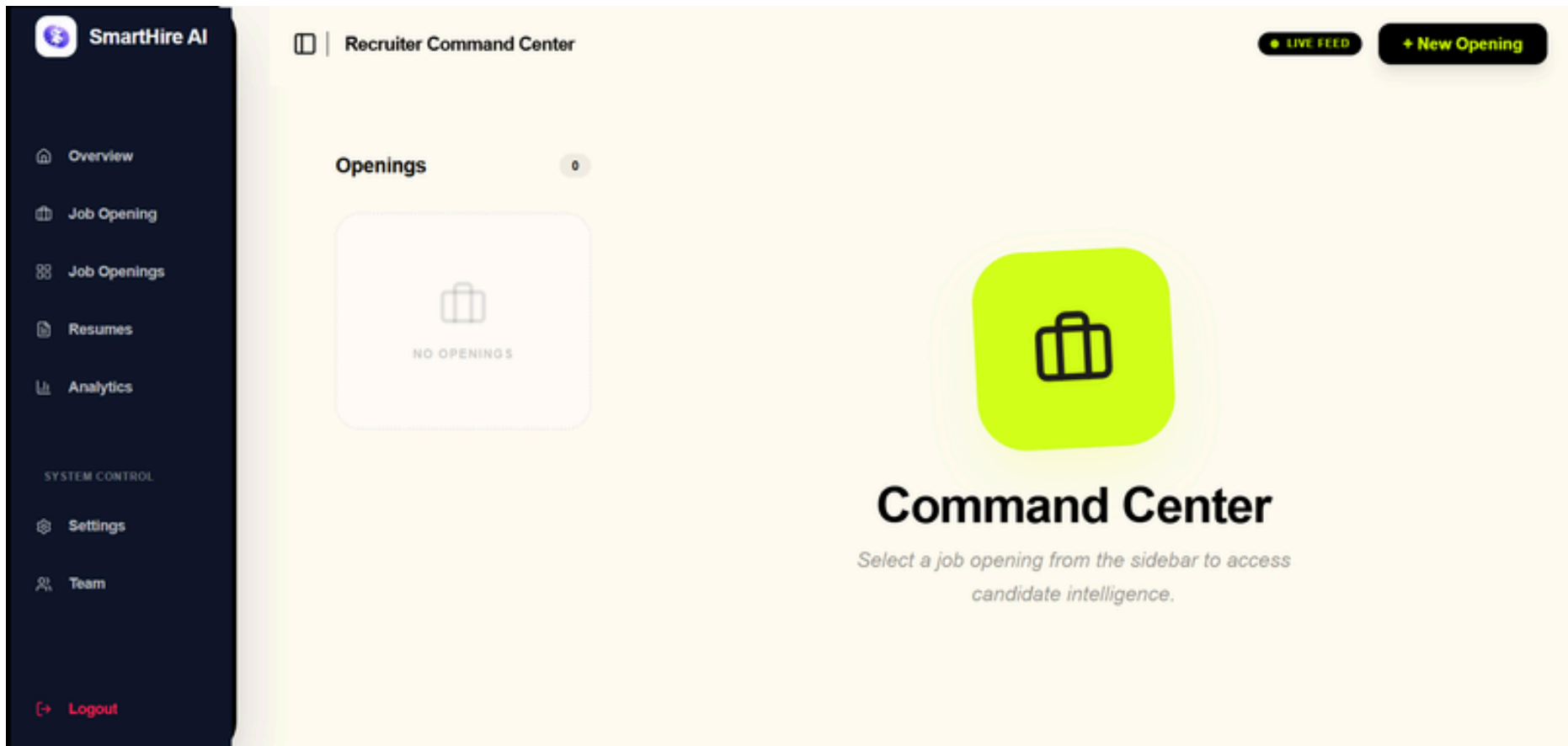


- **Multi-Face Detection**: Uses YOLOv8 to flag if extra persons are in the frame.
- **Attention Tracking**: MediaPipe detects facial orientation to ensure focus on the screen.
- **Browser Lockdown**: Fullscreen enforcement and tab-switching detection prevent external lookups.

Recruiter Insights Dashboard



- Strategic Oversight
- Bulk resume ranking via ChromaDB.
- Detailed AI performance evaluation logs.
- Leaderboards based on semantic fit scores.

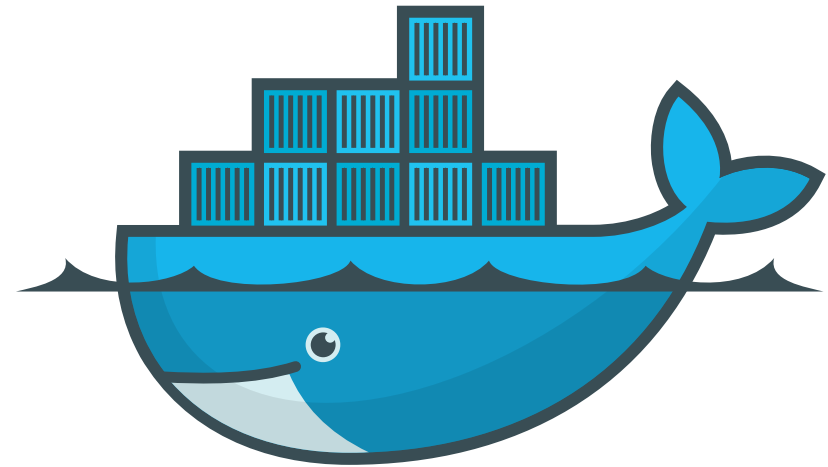


Isolated Execution



Docker Sandboxing

Untrusted candidate code represents a security risk. SmartHire AI executes all code in an ephemeral, strictly controlled environment.



Containerized: Code runs inside ``Dockerfile.sandbox``.

Time Limits: Strict 5-second timeout kills infinite loops.

Zero Network: Containers have no external access.



Interactive Editor

A Monaco-based code editor provides a professional environment for DSA challenges.

- Real-time syntax highlighting.
- Direct terminal feedback from sandbox.



Conclusion & Main Takeaways



Project Summary

SmartHire AI successfully automates the recruitment workflow using AI-powered technologies.

Achievements

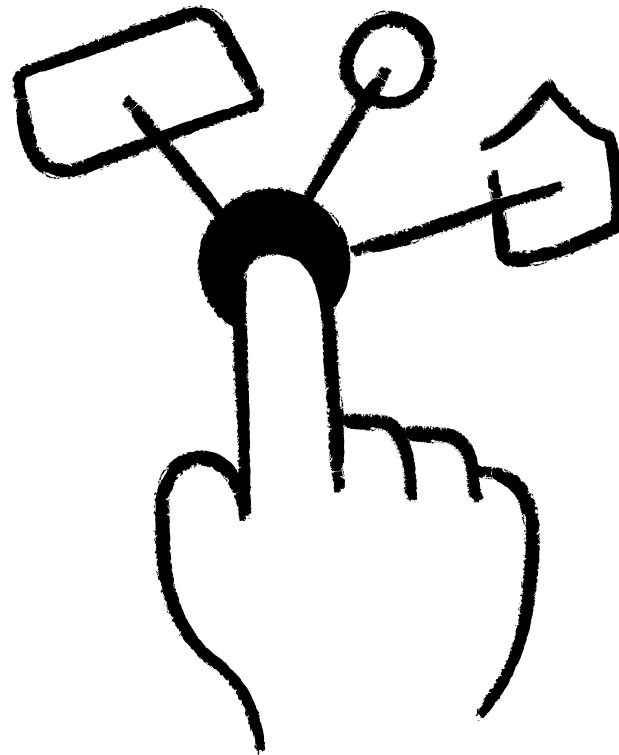
- Automated resume screening
- AI-based interview generation
- Secure coding evaluation
- Real-time candidate monitoring
- Semantic ATS matching

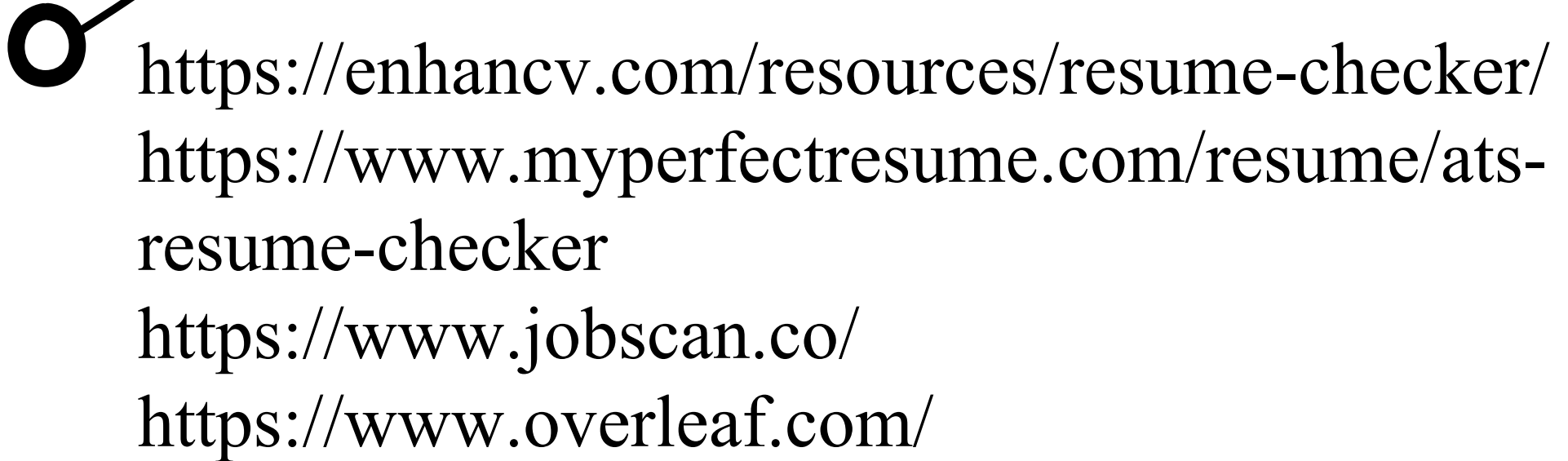
Learning Outcomes

- Full-stack development
- AI integration
- Computer vision implementation
- Docker containerization
- FastAPI backend architecture

Future Scope

- Speech emotion analysis
- Cloud GPU deployment
- Multi-language interview support
- Advanced anti-cheating systems





*Thank
you!*